MOHAMAD RIZK

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EDUCATION

B.S. Cybersecurity & Information Assurance — University of Michigan Dearborn *Expected Dec 2026*
Relevant Coursework: Practical Computer Security, Computer Networks, Introduction to Cryptography | Dean's List (Winter 2024)

TECHNICAL SKILLS

Security: SIEM fundamentals, log, packet, and phishing analysis, incident response, vulnerability assessment, NIST CSF
Security tools: Wireshark, tcpdump, Burp Suite, OpenSSL, Kali Linux, Metasploit (lab)
Programming: Python, JavaScript, Bash, C++, SQL (SQLite)
Web & cloud: Flask, GitHub, Vercel serverless functions, Playwright, AWS fundamentals (IAM, EC2)
ML & data: PyTorch, Hugging Face Transformers, LoRA fine tuning, data pipelines, model evaluation
Systems & network: Linux (Ubuntu, Kali), WSL2, VirtualBox, TCP/IP, DNS, SSL/TLS
Practices: Agile/Scrum, requirements and test planning, client communication, AI assisted development

EXPERIENCE

Cofounder — HR Studios (hr-studios.com) Sep 2026 – Present
- Cofounded a web studio that builds, hosts, and maintains local business sites on monthly plans with ongoing support
- Led client discovery, scoping, and pricing, turning business needs into proposals for websites, apps, and rewards systems
- Built the site and a rate limited serverless contact form with AI assistance; fixed a silent bug so every inquiry arrives
- Run releases through GitHub, Vercel, and Playwright tests with preview review, so nothing untested goes live

Security Analyst Intern — Covision Performance Jun 2025 – Aug 2025
- Evaluated log management, WAF, and bot detection tools against compliance needs to inform tool selection
- Built a risk assessment covering audit deadlines, breach notification, and indemnification terms to show compliance exposure
- Drafted a Vulnerability Disclosure Policy and security.txt so outside researchers can report issues to the trust center

Mobile Repair Technician — Cellphones and Repairs Jan 2024 – Nov 2025
- Diagnosed and fixed hardware and software faults on iOS and Android devices through systematic troubleshooting
- Recovered customer data during operating system repairs, putting data protection first on every job

PROJECTS

Automated Patent Prediction System — Senior Design May 2026 – Dec 2026
- On a 4 person team, built a web app returning a patent's top 3 CPC categories with confidence scores and an audit trail
- Team's models reach 95% top 3 and 79% first guess accuracy across 98 categories on 1,387 unseen applications
- Built the pipeline that parses 20 GB USPTO bulk files into a balanced training set so models don't favor common classes
- Built the Flask and JavaScript front end; added confidence bars and a Classes tab after client testing showed user needs
- Ran client demos as one of two main contacts; wrote the 54 page design document and a 53 case test plan

Membership Inference Attack on a Vision Language Model (Team Lead) Sep 2026
- Led a 3 person team reproducing a 2025 USENIX Security attack that detects whether an image was in LLaVA 7B's training data using only the model's answers, showing how a model can leak what it trained on
- Diagnosed a chance level first run (AUC 0.50) by comparing loss on trained vs. held out images, then retuned LoRA training
- Fixed bugs in the authors' code; final classifier reached 0.72 AUC on single images and 0.98 on groups of 20

LeakLens — Browser Security Extension Jul 2026
- Built a Chrome extension (Manifest V3) that flags Gmail phishing using sender domain and link mismatch indicators
- Integrated Google Safe Browsing via a Cloudflare Worker proxy so the API key never reaches users' browsers
- Added detection for passwords and private data leaking into URLs, where history and logs can expose them

Man in the Middle (MITM) Security Lab Nov 2025
- Built a Lubuntu virtual environment in VirtualBox to simulate application traffic and intercept SSL/TLS communication
- Captured packets with Wireshark and tcpdump to show how TLS misconfigurations expose unencrypted data
- Reproduced certificate validation errors with OpenSSL and published a vulnerability assessment report on GitHub

Android App Static Analysis Lab Apr 2026
- Decompiled and analyzed 10 VPN Android apps, examining AndroidManifest.xml and Network Security Configuration files
- Found cleartext traffic permissions, absent certificate pinning, and NSC misconfigurations across all 10 apps

CERTIFICATIONS

Google Cybersecurity Professional Certificate — Coursera (8 courses: SIEM, incident response, Linux, Python) *Jul 2025*
Scientific Computing with Python — freeCodeCamp (about 300 hours) *Sep 2025*
TryHackMe Pre-Security Learning Path (Linux, networking, web security basics) *Apr 2025*
The Complete C++ Developer Course — Udemy (22.5 hours: OOP, systems programming) *Feb 2025*